Kervyn

AI-Powered PCBA Manufacturing Quality Intelligence Platform

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> *"Transforming complex manufacturing data into actionable, natural-language quality intelligence."*

1. Executive Summary

Modern Printed Circuit Board Assembly (PCBA) environments are highly complex. They generate vast volumes of production and quality data across multiple process stages, machines, and inspection systems (e.g., SMT lines, AOI, ICT). When quality issues arise, identifying the root cause, monitoring production performance, and investigating defects is notoriously difficult because data is siloed.

Kervyn addresses this critical industry challenge. Built on a robust Exasol analytical database and powered by Sarvam AI, Kervyn brings manufacturing traceability, inspection results, and defect monitoring into a single, unified platform. It allows manufacturing engineers and plant managers to ask complex analytical questions in natural language, entirely bypassing the need for manual SQL queries.

2. The Problem Statement

- **Data Fragmentation:** Defect detection, machine logs, and rework histories live in different systems.
- **High Investigation Latency:** Quality engineers spend hours writing complex SQL queries or combining spreadsheets to trace a board's failure back to its origin.
- **Cost of Poor Quality (COPQ):** Without rapid root-cause analysis, production lines continue to manufacture defective units, driving up rework and scrap costs.
- **Inaccessible Analytics:** Floor managers need answers immediately but lack the technical expertise to query massive datasets.

3. The Kervyn Solution

Kervyn is a web-based intelligence platform that seamlessly bridges the gap between raw manufacturing data and human decision-making. By leveraging the high-performance analytics of Exasol and the advanced reasoning of Sarvam AI (sarvam-105b), Kervyn offers:

- **Manufacturing Datasheet:** A centralized dashboard for production KPIs, defect severity analysis, and real-time pass/failure statistics.
- **Kervyn AI (Intelligence Copilot):** A natural-language interface allowing users to converse directly with their manufacturing data.
- **End-to-End Traceability:** A relational data model linking every product, board unit, process stage, and rework event.

4. Key Platform Features

Manufacturing Datasheet

The centralized analytics dashboard retrieves live information directly from the Exasol data warehouse. It provides process-stage filtering, defect-category analysis, production trends, and detailed inspection records. Users can export the complete structured datasheet as JSON for external compliance reporting.

Kervyn AI – Manufacturing Intelligence Copilot

Instead of relying on predefined dashboard widgets, users can ask dynamic, analytical questions such as:

- *"Which manufacturing stage has the highest defect count?"*
- *"Analyse rework costs across production orders."*
- *"Identify machines with unusually high process durations."*

Kervyn AI returns not just text, but structured **Analytical Visualizations** including bar charts, line charts, pie charts, and data tables, alongside the contextually aware AI response.

5. AI Query Architecture & Enterprise Safety

Enterprise data requires strict governance. Kervyn implements a highly structured, secure AI query-planning architecture rather than allowing raw, unrestricted LLM-to-SQL generation.

1. **Query Planning:** Sarvam AI interprets the natural-language question and generates a structured, JSON-based query plan.
2. **Strict Validation:** The backend validates tables, columns, and relationships against an approved schema allowlist.
3. **SQL Compilation:** The validated plan is compiled into secure SQL by the backend.
4. **Bounded Execution:** Queries are limited to 5 tables, 12 fields, and a strict 200-row return limit to protect database performance.

5. **Evidence-Based Response:** Sarvam AI generates the final text analysis grounded strictly in the retrieved database evidence.

6. PCBA Data Traceability Model

The backbone of Kervyn is a highly structured relational database optimized for manufacturing operations:

Entity	Description & Traceability Role
PRODUCT & WORK_ORDER	Links high-level production targets, planned quantities, and product revisions.
BOARD_UNIT	Tracks individual PCB assemblies via unique serial numbers across the shop floor.
PROCESS_STAGE & MACHINE	Maps the physical routing of boards through specific manufacturing equipment.
INSPECTION_RESULT	Records parametric measurements and pass/fail outcomes at automated gates.
DEFECT_EVENT & REWORK	Crucially distinguishes between where a defect is *detected* versus where its origin is *suspected*, tying into rework actions and cost analysis.

7. Technology & Infrastructure Stack

Category	Technology Chosen
Frontend & Visualization	React 19, Vite 8, Recharts (Custom SVG), Lucide React, jsPDF
Backend API Framework	Node.js, Express 5, WebSocket for DB communication
Data Warehouse	Exasol (Deployed on Azure standard instances via Exasol Launcher)
AI Engine	Sarvam AI (Model: sarvam-105b)

8. Future Roadmap & Scaling Scope

While Kervyn currently excels at retrospective quality intelligence and ad-hoc analysis, the architecture is designed to support proactive manufacturing interventions:

- **Predictive Analytics:** Implementing machine learning models to detect early anomalies in machine parameters before defects occur.
- **Real-Time Ingestion:** Streaming IoT data from SMT machines directly into Exasol for sub-second dashboard updates.
- **Automated Root-Cause:** System-generated alerts that automatically trace a spike in defects back to a specific component reel or machine nozzle.
- **Role-Based Access:** Tailored intelligence views for Operators, Quality Engineers, and Plant Directors.

Kervyn — Smarter Manufacturing. Better Quality.